NEXVIA AEGIS V9-M4 Independent Statistical Audit

Phase 5: Audit of Hypothesis H_4 (Task Coverage Floor and Anti-Gaming Guardrail)

Autonomous Verification Subsystem (Google DeepMind Antigravity)

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Abstract

This audit report investigates the statistical design and implementation of Hypothesis H_4 (Task Coverage Floor Precondition) within the NEXVIA AEGIS V9-M4 confirmatory evaluation harness. In selective classification and certified runtime assurance, selective risk can be trivially minimized by policies that abstain excessively. Hypothesis H_4 establishes an operational utility guardrail requiring sample coverage $\hat{C} = M/N \geq C_{\min} = 0.35$ on the uniform mixture holdout $\mathcal{D}_{\text{test}}^{\text{mix}}$ ($N = 1,440$). We audit whether the decision rule operates on the empirical point estimate or an inferential lower confidence bound, analyze exact boundary behavior at $M \in \{503, 504, 505\}$, examine the accompanying two-sided Wilson score confidence interval, and evaluate the anti-gaming classification logic. We verify that the implementation adheres strictly to the preregistered protocol, properly distinguishing an empirical operational precondition from a population-level coverage guarantee.

Keywords: Task Coverage, Selective Abstention, Anti-Gaming Guardrail, Wilson Score Interval, Empirical Threshold Boundary, Runtime Assurance Utility.

1 Executive Summary & Audit Mandate

In safety-critical autonomous systems with an option to abstain, the primary objective is to bound the conditional risk of committed decisions:

$$R_{\text{sel}}(\pi) = \mathbb{P}(\text{FaultyCommitment} \mid \text{Commitment}) \tag{1}$$

However, conditioning on commitment introduces a fundamental failure mode: a degenerate policy that abstains on 99% of episodes can achieve near-zero selective risk simply by avoiding action in all but the most trivial situations.

To prevent this "abstention gaming," Milestone V9-M4 introduces Hypothesis H_4 as an essential operational precondition:

$$\text{Preregistered Rule: } \hat{C}(\hat{\pi}; \mathcal{P}_{12}) = \frac{M}{N} \geq C_{\min} = 0.35 \quad (35.0\%) \tag{2}$$

Phase 5 of the independent audit evaluates:

1. **Decision Rule Specification:** Does the harness evaluate the point estimate $\hat{C} \geq 0.35$, or does it evaluate a lower confidence bound?
2. **Boundary Precision on $N = 1,440$:** How does the decision rule behave at the exact sample boundaries $M = 503$ (34.93%) vs. $M = 504$ (35.00%)?
3. **Wilson Score Interval Construction:** Is the Wilson score interval implemented correctly for descriptive reporting of coverage uncertainty?
4. **Integration with Overall Certification:** Does a failure of H_4 override a nominal pass of H_1, properly labeling the outcome as a degenerate risk-coverage tradeoff?

2 Point Estimate vs. Inferential Lower Confidence Bound

2.1 Protocol Specification

We inspected the governing protocol documents:

- `docs/v9_experiment_protocol.md` Section 3.1.1 (line 22):

  "and empirical holdout coverage $\hat{C}_{test}^{mix}(\hat{\pi}) \geq C_{\min} = 0.35$."

- `docs/v9_m4_readiness_review.md` Section 3 (Table 3, line 56):

  "Hypothesis H_4 | Coverage Floor Precondition | Sample coverage $\hat{C} = M/N$ | $\hat{C}(\hat{\pi}; \mathcal{P}_{12}) \geq C_{\min} = 0.35$ (target ≥ 0.45)."

2.2 Implementation Verification

In `experiments/v9_evaluate_holdout.py`, lines 546–548 implement:

```
# Hypothesis H_4: Coverage Floor Precondition
h4_passed = bool(cov >= 0.35)
h4_status = "PASS_ADEQUATE_UTILITY" if h4_passed else "FAIL_COVERAGE_DEGENERATE"
```

Audit Finding 5.1: The evaluation harness strictly uses the empirical sample proportion `cov = m / n >= 0.35`, exactly reflecting the preregistered protocol. It does **not** silently impose an unapproved lower confidence bound constraint.

3 Boundary Analysis on $N = 1,440$ Holdout Sample

On the uniform mixture holdout $\mathcal{D}_{\text{test}}^{\text{mix}}$, the planned sample size is exactly $N = 1,440$ episodes. The critical integer threshold is:

$$M^* = 0.35 \times 1,440 = 504.0 \text{ episodes} \tag{3}$$

Table 1 documents boundary behavior audited across synthetic sample counts.

Accepted (M)	**Total** (N)	**Empirical Coverage** $\hat{C}$	**Rule** $\hat{C} \geq 0.35$	H_4 **Status**	**Wilson 95% CI**
502	1,440	0.348611 (34.86%)	False	**FAIL**	[0.324431, 0.373598]
503	1,440	0.349306 (34.93%)	False	**FAIL**	[0.325112, 0.374301]
504	1,440	0.350000 (35.00%)	True	**PASS**	[0.325793, 0.375005]
505	1,440	0.350694 (35.07%)	True	**PASS**	[0.326475, 0.375709]
648 (Target)	1,440	0.450000 (45.00%)	True	**PASS**	[0.424472, 0.475795]

Table 1: Boundary evaluation of empirical coverage point estimates and Wilson score intervals.

Audit Verification Finding 5.2:

- At $M = 503$, the coverage is $34.93\% < 35.0\%$, correctly triggering `FAIL_COVERAGE_DEGENERATE`.
- At $M = 504$, the coverage hits exactly 35.00%, achieving `PASS_ADEQUATE_UTILITY`.
- Notice that at $M = 504$, the Wilson lower bound is $0.3258 < 0.35$. Had the harness mistakenly required the Wilson lower bound to exceed 0.35, it would have demanded $M \geq 541$ (37.57%). The code correctly adheres to the registered point-estimate standard.

4 Anti-Gaming Classification Interaction ($H_1 \wedge H_4$)

We inspected how H_4 interacts with H_1 in lines 598–605 of `experiments/v9_evaluate_holdout.py`:

```
# Anti-gaming rule evaluation
if (not h4_passed) or (h1_passed and not h4_passed):
    overall_classification = "CONSERVATIVE_RISK_COVERAGE_TRADEOFF"
elif h1_passed and h4_passed:
    overall_classification = "CERTIFIED_OPERATIONAL_IMPROVEMENT"
else:
    overall_classification = "UNCERTIFIED_POLICY"
```

Audit Finding 5.3 (Anti-Gaming Hierarchy):

- If a candidate policy achieves $M = 60, K = 0$, H_1 passes ($U_{\text{CP},1} \approx 0.0487 \leq 0.05$). However, coverage is only $\hat{C} = 60/1440 \approx 4.17\% < 35\%$.
- The harness refuses to declare `"CERTIFIED_OPERATIONAL_IMPROVEMENT"`.
- Instead, it explicitly classifies the result as `"CONSERVATIVE_RISK_COVERAGE_TRADEOFF"`.
- This confirms that the software architecture enforces the preregistered anti-gaming protocol, preventing trivial over-abstention from masquerading as a certified operational improvement.

5 Conclusion

Phase 5 audit confirms that the implementation of Hypothesis H_4 is mathematically sound, evaluates the exact registered empirical threshold, correctly handles boundary conditions, and prevents anti-gaming vulnerabilities.
Phase 5 Recommendation: `AUDIT_PHASE_5_PASS`

References

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[3] R. G. Newcombe, "Two-sided confidence intervals for the single proportion: comparison of seven methods," *Statistics in Medicine*, vol. 17, no. 8, pp. 857–872, 1998.